You can solve these problems with a DNS name that points to your server. But DNS isn’t fully under your control. DNS resolvers cache responses. DNS servers cache entries, and sometimes they don’t respect your time-to-live (TTL) settings. For example, you might ask DNS servers to only cache the name-to–IP address mapping for one minute, but some DNS servers might use a minimum cache of one day. A better solution is to use a load balancer.

A load balancer can help decouple a system where the requester awaits an immediate response. Instead of exposing your EC2 instances (running web servers) to the outside world, you expose only the load balancer to the outside world. The load balancer then forwards requests to the EC2 instances behind it. Figure 14.1 shows how this works.